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Homework 6

1.

(a)

Show that the distance from the hyperplane $g(\mathbf{x}) = \mathbf{x^T x} + w_0 = 0$ to the point $\mathbf{x}_a$ is $|g(\mathbf{x}_a)|/\|\mathbf{w}\|$ by minimizing $\|\mathbf{x} - \mathbf{x}_a\|^2$ subject to the constraint $g(\mathbf{x}) = 0$.

(b)

Show that the projection of $\mathbf{x}_a$ onto the hyperplane is given by

$$\mathbf{x_p} = \mathbf{x}_a - \frac{g(\mathbf{x}_a)}{\|\mathbf{w}\|^2}\mathbf{w}$$

2.

Let $\mathbf{x_1}, \ldots, \mathbf{x_n}$ be n q-dimensional samples and Q be any nonsingular positive definite $q \times q$ matrix. Show that the vector $\mathbf{x}$ that minimizes

$$\sum_{k=1}^{n} (\mathbf{x_k} - \mathbf{x})^T Q^{-q} (\mathbf{x_k} - \mathbf{x})$$

is the sample mean, $\bar{\mathbf{x}} = \frac{1}{n} \sum_{k=1}^{n} \mathbf{x_k}$.

3.

Consider a linear classifier with discriminant functions $g_i(\mathbf{x}) = \mathbf{w_i^T x} + w_{i0}, i = 1, \ldots, c$. Show that the decision regions are convex by showing that if $\mathbf{x}_1 \in \mathcal{R}_i$ and $\mathbf{x}_2 \in \mathcal{R}_i$ then $\lambda \mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2 \in \mathcal{R}_i$ if $0 \leq \lambda \leq 1$.

By definition, we know that the linear machine will assign $\mathbf{x}$ to class i if $g_i(\mathbf{x}) > g_j(\mathbf{x})$ for all $j \neq i$; as a byproduct of this, we also know that $g_i(\mathbf{x_1}) > g_j(\mathbf{x_1})$ and $g_i(\mathbf{x_2}) > g_j(\mathbf{x_2})$. As suggested by the problem statement, we aim to show that the decision regions are convex by showing that arbitrary point $\lambda\mathbf{x_1} + (1-\lambda)\mathbf{x_2}$ lies within the classification region with $0 \leq \lambda \leq 1$. We substitute this value for $\mathbf{x}$ into the linear classifier discriminant function:

$$g_i(\mathbf{x}) = \mathbf{w_i^T}(\lambda\mathbf{x_1} + (1-\lambda)\mathbf{x_2}) + w_{i0}$$

Looking at this – intuitively, we can see that this essentially is a complementary weighting of the two vectors since they sum to 1. We can therefore redistribute accordingly, with the bias distributed between the λ and $1-\lambda$ weights, since this still adds up to $1 \cdot w_{i0}$:

$$= \lambda(\mathbf{w_i^T x_1} + w_{i0}) + (1-\lambda)(\mathbf{w_i^T x_2} + w_{i0})$$

and then substituting in the linear discriminant function definition accordingly:

$$g_i(\mathbf{x}) = \lambda g_i(\mathbf{x_1}) + (1-\lambda) g_i(\mathbf{x_2})$$

Based on the inequalities we previously established, we also know that $\lambda g_i(\mathbf{x_1}) \geq \lambda g_j(\mathbf{x_1})$, and $(1-\lambda) g_i(\mathbf{x_2}) \geq (1-\lambda) g_i(\mathbf{x_2})$, since multiplying by the $0 \leq \lambda \leq 1$ doesn't change the sign of the inequality but introduces the potential of possible equality in the case that either λ or $1-\lambda$ is 0. Combining this with the actual discriminant function that we found for $\mathbf{x}$, we see that we have

$$\lambda g_i(\mathbf{x_1}) + (1-\lambda) g_i(\mathbf{x_2}) > \lambda g_j(\mathbf{x_1}) + (1-\lambda) g_j(\mathbf{x_2})$$

with this being a strict inequality since only one of the weights being able to be 0, which is the only case of equality, thus leaving the other term to be a strict inequality. We already solved for this previously, so we are essentially just left with

$$g_i(\mathbf{x}) > g_j(\mathbf{x})$$

which shows us that this intermediate point satisfies the linear machine assignment rule that we first started with, which therefore indicates that indeed $\mathbf{x} \in \mathcal{R}_i$ as we hoped to show.

4.

In the gradient descent algorithm $\mathbf{a}_{k+1}$ is obtained from $\mathbf{a}_k$ by

5.

Show that the partial derivatives of the functions $y_i = \exp(a_i)/\sum_k \exp(a_k)$ used in multiple class logistic discrimination are given by

$$\frac{\partial y_i}{\partial a_j} = y_i(\delta_{ij} - y_j)$$

where $\delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & \text{otherwise.} \end{cases}$

We have that $y_i = \frac{\exp(a_i)}{\sum_k \exp(a_k)}$. Generally, we use the quotient rule here:

$$\frac{\partial y_i}{\partial a_j} = \frac{(\frac{\partial}{\partial a_j}\exp(a_i)) \cdot (\sum_k \exp(a_k)) - \exp(a_i) \cdot \frac{\partial}{\partial a_j}\sum_k \exp(a_k)}{(\sum_k \exp(a_k))^2}$$

Now let's start case by case for the value of δ_{ij}. For $i \neq j$:

$$\begin{aligned} \frac{\partial y_i}{\partial a_j} &= \frac{0 \cdot (\sum_k \exp(a_k)) - \exp(a_i) \cdot \frac{\partial}{\partial a_j}\sum_k \exp(a_k)}{(\sum_k \exp(a_k))^2} \\ &= \frac{-\exp(a_i) \cdot \frac{\partial}{\partial a_j}\sum_k \exp(a_k)}{(\sum_k \exp(a_k))^2} \\ &= \frac{-\exp(a_i) \cdot \exp(a_j)}{(\sum_k \exp(a_k))^2} \\ &= \frac{-\exp(a_i)}{\sum_k \exp(a_k)} \cdot \frac{\exp(a_j)}{\sum_k \exp(a_k)} \end{aligned}$$

and substituting back based on the form of y_i that we had before:

$$= -y_i y_j$$

$$\frac{\partial y_i}{\partial a_j} = y_i(0 - y_j)$$

which indeed fits the equation we aimed to find with the corresponding value of $\delta_{ij} = 0$. Now solving for when $i = j$:

$$\frac{\partial y_i}{\partial a_j} = \frac{\exp(a_i) \cdot (\sum_k \exp(a_k)) - \exp(a_i) \cdot \frac{\partial}{\partial a_j} \sum_k \exp(a_k)}{(\sum_k \exp(a_k))^2}$$

$$= \frac{\exp(a_i) \cdot (\sum_k \exp(a_k)) - \exp(a_i) \cdot \exp(a_j)}{(\sum_k \exp(a_k))^2}$$

$$= \frac{\exp(a_i) \cdot (\sum_k \exp(a_k))}{(\sum_k \exp(a_k))^2} - \frac{\exp(a_i) \cdot \exp(a_j)}{\sum_k \exp(a_k) \cdot \sum_k \exp(a_k)}$$

$$= \frac{\exp(a_i)}{\sum_k \exp(a_k)} - \frac{\exp(a_i)}{\sum_k \exp(a_k)} \cdot \frac{\exp(a_j)}{\sum_k \exp(a_k)}$$

and doing the same substitutions:

$$= y_i - y_i \cdot y_j$$

$$\frac{\partial y_i}{\partial a_j} = y_i(1 - y_j)$$

which also fits the equation we aimed to find with corresponding value $\delta_{ij} = 1$.

6.

In Module 1 we introduced nine (9) datasets. The data files are posted here in Module 9. Choose three (3) of these datasets and for each of the three:

(a)

i.

ii.

(a)

i.

ii.

iii.

iv.

A.